

Qatar-2b

R-2232

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140701 · created 2026-08-15T01:20:06+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456839.706271 +/- 5.7e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.224 +/- 0.025
Transit depth (Rp/Rs)^2	5.03 +/- 1.12 [%]
Orbital Inclination (inc)	84.54 +/- 1.5
Airmass coefficient 1 (a1)	1.378 +/- 0.036
Airmass coefficient 2 (a2)	-0.131 +/- 0.012
Scatter in the residuals of the lightcurve fit is	2.6 %
Best Comparison Star	None
Optimal Aperture	3.42
Optimal Annulus	16.41
Transit Duration (day)	0.0646 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.224 $\pm$ 0.025 vs 0.16327 (deviation 2.43 σ)
Duration fit vs published	0.0646 d vs 0.0767 d (deviation 2.68 σ)
Mid-time O-C	0.3 min
Depth SNR	9.0
Residual scatter	2.6 %

Light curve

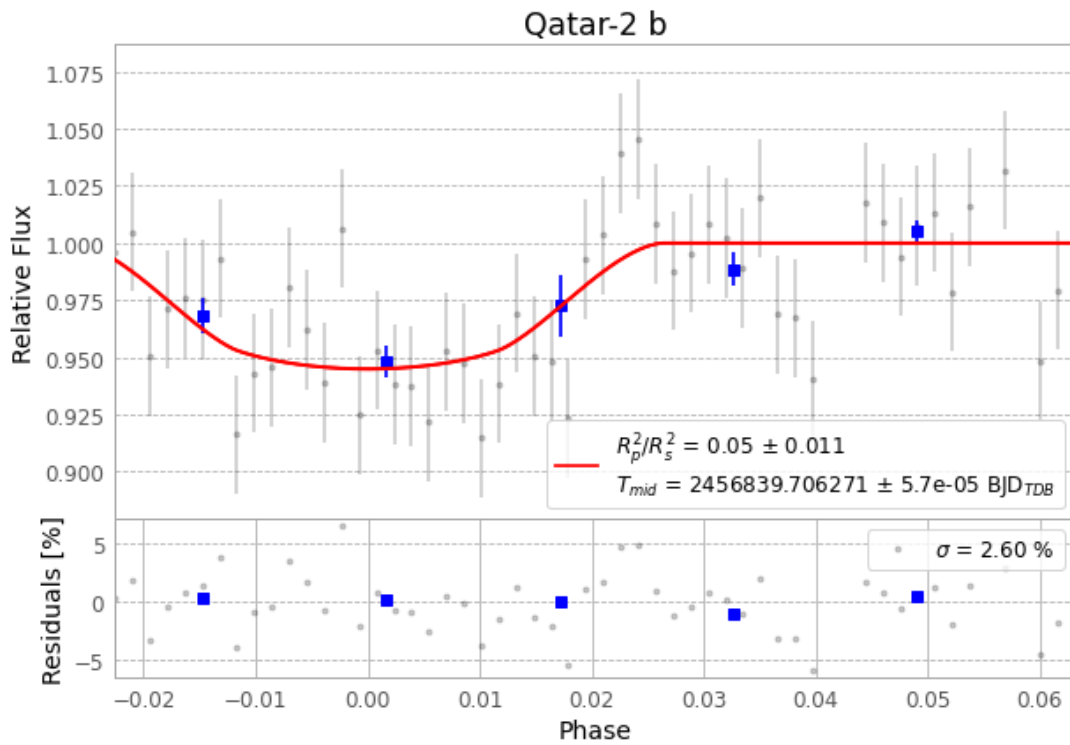


Figure 1 — FinalLightCurve_Qatar-2 b_2014-06-30.png (SHA-256 dc6e918006df82d1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0bcc026e7b648e53...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

56 FITS frames (37 MB), Qatar - 2140701040913.FITS → Qatar - 2140701065413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140701.log (4418d685e673...), FinalLightCurve_Qatar-2 b_2014-06-30.png (dc6e918006df...), FinalLightCurve_Qatar-2 b_2014-06-30.csv (29ebf5eb3078...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:20Z.